ASHWANI KUMAR DWIVEDI

Senior Android Engineer | Modern Android (Compose) & Scalable Mobile Architecture

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• Noida, India

Summary

Senior Android Engineer with **5.5 years** of experience building **high-scale, production-grade mobile applications** and core system components powering millions of flagship devices. Deep expertise in **native Android** development across both modern **Jetpack Compose** architectures and **legacy XML/View systems**, with a strong focus on **Clean Architecture, MVVM, and reactive state management** using **Kotlin Coroutines and StateFlow.** Proven track record of optimizing **UI rendering performance (60 FPS), eliminating memory leaks,** and driving reliability initiatives that reduced production defects by **30%.** Adept at leading cross-functional platform initiatives, establishing **automated CI/CD testing pipelines**, and mentoring engineering teams to ship scalable, maintainable mobile codebases.

Experience

Samsung R&D Institute India

Noida

Software Engineer

02/2021 - 07/2026

- **UI & Frame Rate Optimization:** Profiled view rendering using Android Studio Profiler, Systrace, and Perfetto, eliminating main thread bottlenecks to maintain smooth 60 FPS performance and reduce jank by 35%.
- **System Component Scalability:** Scaled and maintained core Android components across multiple flagship device lines, delivering 400+ production change lists with high modularity and zero downtime.
- **Production Reliability & Bug Remediation:** Resolved 500+ high-severity production defects and edge-case application crashes across millions of active devices, driving down post-release platform error rates by 30%.
- **CI/CD & Automated Testing Infrastructure:** Architected end-to-end testing suites spanning unit, UI (Espresso), and integration tests, reducing CI pipeline turnaround time by 50% while safeguarding backward compatibility across multiple Android OS versions.
- **Code Quality & Technical Mentorship:** Led architectural design discussions for 20+ major platform features; instituted strict code review standards and modern development guidelines across a 15-engineer team to maintain robust design patterns and high code quality.
- **Cross-Functional Collaboration:** Partnered closely with cross-functional product, QA, and security teams, authoring technical specs and onboarding documentation to accelerate feature delivery cycles.

Projects

Fluid Vector AR | Kotlin, Jetpack Compose, ARCore, CameraX, Room DB

An **advanced vector drawing studio** built with **Jetpack Compose** and **Google ARCore**, featuring **real-time Bezier curve smoothing**, **live CameraX stencil tracing**, and **3D spatial projection**.

- Engineered a **high-performance vector drawing** & **AR stencil studio** in **100% Kotlin** using **Jetpack Compose, Clean Architecture, and MVVM.**
- Implemented an **infinite 2D canvas** with **sub-pixel Bezier curve smoothing** and **matrix transformations**, maintaining **60 FPS** rendering under heavy stroke loads.
- Built an interactive shape engine featuring **vertex-level polygon deformation**, **custom undo/redo action stacks**, and **multi-format export pipelines (SVG, PNG, JPEG)**.
- **Integrated Google ARCore/SceneView** for **3-axis spatial projection** and **CameraX live preview overlay** with **hardware-accelerated opacity blending** for precision tracing.
- Developed **local persistence** using **Room DB** & **DataStore Preferences**, coupled with **Firebase Auth** and **Android Credential Manager API** for seamless authentication.

iOS Health Application | SwiftUI, JavaScript

Architected and integrated hybrid **Web-to-Native iOS** features using **SwiftUI**, custom **JavaScript** bridging, and **deep-linking pipelines** for the Health application.

- **Engineered a high-performance WebView module** in SwiftUI with backward compatibility for older iOS versions, maintaining a smooth **60 FPS rendering speed** and clean architecture.
- **Applied secure JavaScript bridging and deep-linking protocols** to enable seamless bidirectional communication and direct navigation between web components and native screens, reducing data latency by **30%**.
- **Optimized hybrid view performance and memory utilization**, cutting frame drops by **25%** and improving responsiveness across lower-end devices.
- **Standardized web-to-native integration workflows**, accelerating feature deployment cycles by **20%** for the iOS Health platform.

Education

Birla Institute of Technology Pilani

Pilani

Masters of Technology

04/2022 - 03/2024

National Institute of Technology Sikkim

Sikkim

Bachelors of Technology

08/2016 - 03/2020

Skills

Java · Kotlin · Swift · Python · C/C++

Android · Jetpack Compose · Android SDK · Coroutines · StateFlow · Hilt · Room DB · CameraX · ARCore · SwiftUI

MVVM · Clean Architecture · MVVM · System Design · Android Profiler · Systrace · Perfetto · Git · CI/CD